

Week 2- Exercise – Individual - Data Quality Check with a FAIR Dataset Using Tableau Prep

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Dataset Description

The WHO Global Health Observatory dataset (2019) compiles annual average air pollution levels—PM2.5, PM10, and NO2 ($\mu\text{g}/\text{m}^3$)—for 50 major cities across 30 countries, alongside contextual variables such as country, population, WHO region, data source, and last update, providing a standardized snapshot of urban air quality for global comparison.

Identified Data Quality Issues

- **Duplicate records** were the first issue spotted — London (UK) and Berlin (Germany) each appeared twice with identical values, adding 2 extra rows to what should have been a 50-row dataset.
- **Missing values** affected several pollutant fields. PM10 was absent for five cities including Kolkata, Kano, and Brasília, while PM2.5 and NO2 were each missing for one city.
- **Inconsistent capitalisation** appeared in the Country field, where 'indonesia' and 'philippines' were stored in lowercase while every other country used title case.
- **Redundant fields** were also present 'Data Source' and 'Last Updated' held the same value in every row ('WHO' and '06/15/2021' respectively), making them analytically useless.
- **Non-standard field names** containing special characters (μ , /) and spaces made the dataset harder to work with in downstream tools like Python or SQL.

Process in Tableau Prep (explanation of the cleaning steps)

The entire workflow was built in a Clean step within Tableau Prep, keeping all transformations transparent and traceable through the Changes panel.

- Duplicate city-year records were identified using a calculated field ("Is Duplicate Row?") and the repeated entries for London and Berlin were removed.
- Redundant columns ('Data Source' and 'Last Updated') were dropped since they contained constant values across all rows.
- Field names were standardised to remove special characters and spaces (e.g. "PM2.5 ($\mu\text{g}/\text{m}^3$)" → "PM2_5_ug_m3") to ensure compatibility with tools like SQL and Python.
- Calculated fields were introduced to flag missing values (e.g. IF ISNULL([PM2_5_ug_m3]) THEN 'Missing' ELSE 'Available' END), preserving incomplete records while making gaps explicit.
- The cleaned dataset was then exported as **output.csv**.

Screenshots of the workflow

The top screenshot shows the 'Clean 1' step of a data workflow. The interface includes a 'Changes (12)' list on the left, a 'Filter' section, and a data table. The 'Changes' list includes: 'Remove Field' (Last Updated), 'Rename Field' (WHO_Region), 'Remove Field' (Data Source), 'Calculated Field' (Is Duplicate Row?), 'Filter' (Is Duplicate Row?), 'Rename Field' (PM2_5 ug_m3), 'Rename Field' (PM10 ug_m3), 'Rename Field' (NO2 ug_m3), 'Calculated Field' (PM10 ug_m3 Missing), and 'Merge Fields' (WHO_Region). The data table has columns: WHO_Region, Country, City, Year, PM2_5 ug_m3, PM2_5 Missing, PM10 ug_m3, PM10 ug_m3 Missing, NO2 ug_m3, and Population. The bottom screenshot shows the 'Output' step of the workflow. The interface includes a 'Save output to' section, a 'Write Options' section, and a data table. The 'Save output to' section includes a 'Browse' button and a 'File' input. The 'Write Options' section includes a 'Full refresh' checkbox and a 'Create table' checkbox. The data table has columns: WHO_Region, Country, City, Year, PM2_5 ug_m3, PM2_5 Missing, PM10 ug_m3, PM10 ug_m3 Missing, NO2 ug_m3, and Population.

WHO_Region	Country	City	Year	PM2_5 ug_m3	PM2_5 Missing	PM10 ug_m3	PM10 ug_m3 Missing	NO2 ug_m3	Population
EUR	Netherlands	Amsterdam	2,019	10.2	Available	17.8	Available	29.4	872,680
SEAR	Thailand	Bangkok	2,019	26.3	Available	48.4	Available	41.2	10,539,000
WPR	China	Beijing	2,019	42.1	Available	68.4	Available	48.3	21,893,095
EUR	Germany	Berlin	2,019	12.4	Available	18.2	Available	28.1	3,660,491
AMR	Colombia	Bogotá	2,019	17.8	Available	33.4	Available	36.2	7,743,905
AMR	Brazil	Brasilia	2,019	nul	Missing	nul	Missing	22.3	3,033,268
AMR	Argentina	Buenos Aires	2,019	14.8	Available	28.6	Available	35.4	15,584,428
EMR	Egypt	Cairo	2,019	92.3	Available	185.4	Available	51.2	20,076,000
AMR	USA	Chicago	2,019	8.4	Available	16.9	Available	26.4	2,693,976
EUR	India	Delhi	2,019	98.6	Available	226	Available	53.4	32,941,309
SEAR	Bangladesh	Dhaka	2,019	97.3	Available	203.4	Available	52.1	21,005,860
EUR	Germany	Hamburg	2,019	11.8	Available	17.9	Available	27.3	1,841,179
SEAR	Vietnam	Ho Chi Minh City	2,019	28.4	Available	nul	Missing	38.4	9,077,158
EUR	Turkey	Istanbul	2,019	25.6	Available	46.8	Available	40.3	15,462,452
WPR	Indonesia	Jakarta	2,019	39.2	Available	75.4	Available	43.1	10,562,000
AFR	South Africa	Johannesburg	2,019	33.4	Available	62.1	Available	38.4	5,635,127
AFR	Nigeria	Kano	2,019	82.1	Available	nul	Missing	42.3	3,828,709
EMR	Pakistan	Karachi	2,019	75.2	Available	nul	Missing	46.3	16,093,786
SEAR	India	Kolkata	2,019	55.2	Available	nul	Missing	38.6	14,974,073
AFR	Nigeria	Lagos	2,019	78.4	Available	163.2	Available	45.2	14,862,111

WHO_Region	Country	City	Year	PM2_5 ug_m3	PM2_5 Missing	PM10 ug_m3	PM10 ug_m3 Missing	NO2 ug_m3	Population
AMR	Brazil	São Paulo	2,019	15.3	Available	31.2	Available	34.7	12,325,232
EUR	Germany	Hamburg	2,019	11.8	Available	17.9	Available	27.3	1,841,179
AMR	Brazil	Rio de Janeiro	2,019	18.7	Available	36.4	Available	29.4	6,747,815
WPR	China	Shanghai	2,019	35.2	Available	57.8	Available	44.1	24,870,895
EUR	Netherlands	Amsterdam	2,019	10.2	Available	17.8	Available	29.4	872,680
SEAR	India	Delhi	2,019	98.6	Available	226	Available	53.4	32,941,309
SEAR	Thailand	Bangkok	2,019	26.3	Available	48.4	Available	41.2	10,539,000
EUR	France	Paris	2,019	14.2	Available	21.6	Available	31.8	2,141,000
AMR	Canada	Vancouver	2,019	5.9	Available	11.8	Available	18.7	676,218
AMR	USA	Chicago	2,019	8.4	Available	16.9	Available	26.4	2,693,976
EUR	Russia	Moscow	2,019	16.7	Available	28.4	Available	35.2	12,006,468
SEAR	Thailand	Bangkok	2,019	26.3	Available	48.4	Available	41.2	10,539,000
EUR	Spain	Madrid	2,019	9.8	Available	18.4	Available	40.6	3,305,408
AMR	Canada	Toronto	2,019	7.4	Available	14.2	Available	25.3	2,731,571
AMR	USA	Los Angeles	2,019	11.2	Available	22.4	Available	32.1	3,978,576
SEAR	Vietnam	Ho Chi Minh City	2,019	28.4	Available	nul	Missing	38.4	9,077,158
EUR	Poland	Warsaw	2,019	22.7	Available	41.3	Available	36.8	1,794,166
WPR	Australia	Melbourne	2,019	5.2	Available	11.8	Available	18.9	5,078,000
WPR	Australia	Sydney	2,019	5.8	Available	12.6	Available	20.3	5,312,000
EMR	Pakistan	Karachi	2,019	75.2	Available	nul	Missing	46.3	16,093,786
WPR	Japan	Tokyo	2,019	11.7	Available	20.1	Available	22.4	13,960,000
EMR	Pakistan	Lahore	2,019	66.1	Available	179.3	Available	44.2	13,095,166
SEAR	Bangladesh	Dhaka	2,019	97.3	Available	203.4	Available	52.1	21,005,860
AMR	Mexico	Mexico City	2,019	20.3	Available	42.1	Available	51.3	9,209,944
EUR	Germany	Munich	2,019	13.1	Available	19.8	Available	30.5	1,484,226
WPR	China	Shenzhen	2,019	22.3	Available	43.6	Available	nul	13,438,202
AMR	Argentina	Buenos Aires	2,019	14.8	Available	28.6	Available	35.4	15,584,428
AMR	USA	New York	2,019	6.9	Available	14.8	Available	28.7	8,336,817
AFR	Nigeria	Lagos	2,019	78.4	Available	163.2	Available	45.2	14,862,111
AFR	Nigeria	Kano	2,019	82.1	Available	nul	Missing	42.3	3,828,709
EMR	Saudi Arabia	Riyadh	2,019	102.4	Available	214.6	Available	33.2	7,876,654
EMR	Iran	Tehran	2,019	36.8	Available	72.4	Available	48.3	9,275,009
EUR	Turkey	Istanbul	2,019	25.6	Available	46.8	Available	40.3	15,462,452

Conclusion

In conclusion, the cleaning process significantly improved the dataset's alignment with FAIR (Findable, Accessible, Interoperable, Reusable) principles. By removing duplicates and redundant fields, the dataset became more reliable and easier to interpret, enhancing its reusability. Standardising field names and formats improved interoperability. The addition of missing value flags preserved transparency while maintaining accessibility to incomplete data rather than discarding it.